

Shubham Bhatt

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Skills

Toolkits & Frameworks: Jetpack Compose, SwiftUI

Android: Kotlin, Coroutines, Flow, Structured Concurrency, Hilt DI, Room, WorkManager, Navigation Component

iOS: Swift, iOS 17+ Observable, OperationQueue, URLSession, Combine, Live Activities / Dynamic Island, SwiftData

Architecture: MVVM, Clean Architecture, Unidirectional Data Flow (UDF), Modularization, Reactive Systems

Networking & Data: REST APIs, WebSockets, Retrofit, AES-256 Encryption

Tools & Practices: Android Studio, Debugging, Profiler, Git, Microsoft Clarity, CI/CD, Firebase, Crashlytics

Work Experience

Native Mobile Engineer — WeHear App

2023 – Present

- **Led end-to-end architecture and delivery of WeHear** — a production-grade wearable companion platform encompassing real-time BLE device control, 100+ language live translation, AI-powered meeting notes and summaries across Android (Kotlin, Coroutines, Flow, Jetpack Compose, Hilt, Clean Architecture) and iOS (Swift, SwiftUI, Observable, OperationQueue), shipping full cross-platform feature parity across 2 production releases.
- **Collaborated on-site with firmware engineers in China** to co-design and iterate BLE communication protocols across 5+ firmware versions — defining 42+ command types, packet structures, acknowledgment flows, and error recovery contracts, ensuring deterministic alignment between Android application layer and embedded firmware constraints.
- Engineered a **robust BLE + OTA firmware pipeline** with serialized command execution, acknowledgment-based flow control, and structured failure recovery — improving firmware update success rate from **~65% to 98.5%** and reducing average update completion time by **~40%**.
- Built **background sync and offline-resilient message delivery** using WorkManager with constraint-based scheduling and exponential backoff, ensuring **zero message loss** during connectivity drops; secured all health and conversation data with **AES-256 encryption** meeting data sensitivity requirements.
- Designed **reactive data pipelines** using Kotlin Flow + structured concurrency for deterministic state propagation across GATT, WebSocket, and UI layers — reducing ANR rate by **~70%** by eliminating main-thread blocking across BLE callbacks, and cutting unnecessary Compose recompositions by **~60%**.
- Built **real-time translation** (100+ languages) over WebSockets achieving **<2s end-to-end latency**; gathered and synthesized user feedbacks during on-site China visits, driving **3+ feature iterations** that improved language recognition accuracy to **85% to 90%**; collaborated with product design and firmware teams across the full **PRD-to-production** lifecycle.
- **Invented PairHub™** — an intelligent multi-device switching system enabling seamless transition between previously connected devices without manual Bluetooth re-pairing, delivering a frictionless ecosystem-like connectivity experience — supporting up to **4 simultaneously paired wearables** and improving device switch latency by **~55%**.
- Delivered **Personal Fit – a Hearing Test-driven** adaptive audio personalization system performing real-time EQ tuning improving personalized hearing experience by **~85%** based on post-fitting user satisfaction benchmarks.

NuConnect — BLE Companion Platform for Smart Hearing Aids

Project

- **Architected NuConnect to accommodate 5+ hearing aid device families** — designing a model-driven, capability-abstracted architecture (MVVM, Hilt, Room) where device-specific features, firmware variants, and hardware capabilities are resolved at runtime, enabling a single codebase to scale across an entire product family; led end-to-end development across Android (Kotlin, Flow) and iOS (Swift, SwiftUI, Observable), owning GATT communication, device pairing, command serialization, state synchronization, and full release delivery.
- Built **Config Assistant** — an intelligent multi-channel hearing calibration engine that tunes **16+ DSP channels** (beyond standard EQ) through an environment-aware Q&A flow; normalizes user-reported acoustic issues such as sharp whistle feedback via targeted **channel suppression** — **reducing in-clinic audiologist visits** by **~30%** and cutting average fitting time from **45 minutes to under 15 minutes**.
- Delivered **Remote Fitting** — cloud-configured device tuning profiles applied in real time via GATT writes — enabling audiologist-prescribed sound profiles to reach patients remotely without physical clinic visits, reducing revisit rates by **~25%**.

- Engineered a **low-latency BLE communication layer** with serialized command queuing, device-specific routing, and structured response parsing — reducing BLE communication latency by **~70%**; co-designed the underlying protocol on-site with firmware engineers, cutting device discovery and pairing time from **10s to 4s** and achieving **99.2%** command acknowledgment reliability.
- Implemented **dynamic multi-device support** via runtime device detection, command abstraction, and parser-based feature routing — enabling a single app binary to support **6+ hardware models** and multiple firmware variants without any platform-level changes.

Native Mobile Intern — Simform Solutions LLP

Jan 2023 – Jun 2023

- Developed Android features using Kotlin with MVVM architecture, Retrofit for REST networking, structured navigation, and Room-based pagination caching — reducing data load times by **~40%** on large dataset screens through a cache-first architecture.
- Built iOS modules using UIKit, MVVM + Coordinator pattern, URLSession networking, and CoreData-backed offline caching; contributed to cross-platform feature delivery and code reviews in an agile team environment.

Education

L.D. College of Engineering

2019 – 2023

Bachelor of Engineering

CGPA: 8.75